

Docket No.: A.24-03-019

Exhibit No.: _____

Date: January 8, 2025

Witness: Zach Woogen

**DIRECT TESTIMONY OF ZACH WOOGEN
ON BEHALF OF THE VEHICLE-GRID INTEGRATION COUNCIL**

1 **I. INTRODUCTION**

2 **Q. Please state your name, title, and business address.**

3 A. My name is Zach Woogen. I am the Interim Executive Director for the Vehicle-Grid
4 Integration Council (“VGIC”). My business address is 1401 21st St, Suite 5409,
5 Sacramento, California 95811.

6 **Q. On whose behalf are you testifying?**

7 A. I am testifying on behalf of VGIC.

8 **Q. What is VGIC?**

9 A. VGIC is a 501(c)6 membership-based trade association committed to advancing the role
10 of electric vehicles (“EV”) and vehicle-grid integration (“VGI”) through policy
11 development, education, outreach, and research. VGIC supports the transition to a
12 decarbonized transportation and electric sector by ensuring the value from EV
13 deployments and flexible EV charging and discharging is recognized and compensated in
14 support of achieving a more reliable, affordable, and efficient electric grid.

15 **Q. Who are VGIC’s current members?**

16 A. VGIC’s members include BorgWarner, Fermata Energy, Ford Motor Company, General
17 Motors, American Honda Motor Company, Nissan Group of North America,
18 Bidirectional Energy, dcbel, Eaton, Emporia, EnergyHub, ev.energy, Flux, Gravity,
19 GridBeyond, GridWiz, Hyundai Cradle, Kaluza, Landis and Gyr, Leap, Mercedes-Benz,
20 Orange Charger, Peak Power, Powerflex, QCells, Stellantis, Sumitomo Electric, Sunrun,
21 Tesla, The Mobility House, Toyota Motors North America, WeaveGrid, Clean Power

1 Alliance, Holy Cross Energy, Los Angeles Department of Water & Power, Peninsula
2 Clean Energy, and Sacramento Municipal Utility District.¹

3 **Q. Please summarize your professional background and qualifications.**

4 A. I am a member of VGIC's leadership team where my primary responsibilities are
5 managing policy advocacy and regulatory engagement, directing research and education,
6 and overseeing day-to-day functions of the organization. VGIC's team is globally
7 recognized for its expertise on VGI topics, including issues related to managed EV
8 charging platforms, bidirectional EV charging solutions, and EV charging technologies
9 deployed alongside other distributed energy resources. Before joining VGIC in 2024, I
10 was a Senior Manager within Strategen's consulting practice, where one of my
11 responsibilities included managing the VGIC, which, at the time, was one of the firm's
12 primary clients. In addition to VGIC, I managed Strategen's other projects within the
13 VGI practice and managed or supported projects for numerous client engagements related
14 to resource planning, renewable energy, energy storage, utility rate design and program
15 design, and utility business models and strategy. I have a Bachelor of Science degree in
16 Environmental Economics and Policy from the University of California, Berkeley.

17 **Q. What is the purpose of your testimony?**

18 A. The purpose of my testimony is to respond to Southern California Edison's ("SCE")
19 Vehicle-to-Grid Resource Proposal ("VGRP"), an export compensation mechanism for
20 customers with bidirectional charging solutions, as filed in A.24-03-019, as well as the

¹ The opinions expressed in this testimony reflect those of VGIC, and do not necessarily reflect the views of all of the individual VGIC member companies.

Public Advocates Office’s (“Cal Advocates”) prepared testimony related to SCE’s proposed VGRP. Specifically, I will address the need for an effective static export credit for customers with bidirectional charging systems. I will also address deficiencies in SCE’s proposal and Cal Advocates’ testimony. In doing so, I propose modifications to SCE’s proposed export compensation mechanism for VGRP customers.

II. BACKGROUND ON CURRENT AND NEAR FUTURE POTENTIAL FOR VGI TO SUPPORT RATEPAYER AFFORDABILITY GOALS AS WELL AS GRID RELIABILITY DURING NET PEAK LOAD HOURS.

Q. SCE’s Amended Rate Design Proposals details that “Despite their considerable energy storage potential, EVs actively engage in transportation services for a fraction of the day and can remain idle for a significant portion of each day. This underutilization of available energy in mobile-battery assets underscores a latent resource that, if tapped effectively, could help address the burgeoning challenges facing the grid as it integrates increasing volumes of renewable energy sources.” Does VGIC agree?

A. Yes. VGIC agrees that EVs are underutilized as a battery energy storage resource and represent untapped potential to support the integration of renewable energy. As such, VGI efforts broadly represent a significant opportunity to both advance the VGI industry and deliver much needed grid benefits. There is no doubt that EVs on the road today, plus those that will be by the anticipated by the 2027 launch of VGRP, have the technical capability to meaningfully support the integration of renewable energy resources through a mix of unidirectional (“V1G”) and bidirectional vehicle-to-everything (“V2X”) charging activities. However, market incentives to support this integration of renewable

energy through opportunities like that proposed in SCE's VGRP have been slower to develop.

Q. SCE's Amended Rate Design Proposals concludes by stating: "By incentivizing customer participation in V2G, *grid-as-a-service* exports and providing a transparent compensation mechanism, SCE paves the way for a sustainable and resilient energy future while empowering customers to actively contribute to grid decarbonization efforts and manage their energy bills." Does VGIC agree?

A. Yes. VGIC agrees that in addition to the renewable energy integration benefit detailed above, incentivizing customers to participate in V2G exports supports a resilient energy future in which customers are empowered to contribute to grid decarbonization efforts and manage their energy bills.

Q. Can EVs support other important policy goals, including affordability for all ratepayers – not just EV drivers – and increased system reliability?

A. Yes. VGI strategies, including V1G managed charging and V2X bidirectional charging, if designed appropriately, can reduce costs to all ratepayers, including EV owners and non-EV customers. By supporting the more efficient use of existing grid infrastructure and accelerating transportation electrification (and the resulting kWh sales), average rates for all customers can be lower than they would be without effective VGI strategies. Furthermore, EVs can meaningfully reduce net peak load through both V1G managed charging and V2X bidirectional charging activities. This is a critical feature of VGI as California faces increased extreme weather events and resulting emergency reliability conditions.

1 **Q. Is VGIC confident that EVs can provide a meaningful contribution, in MW terms, if**
2 **encouraged to do so through appropriate export credits for bidirectional charging**
3 **customers?**

4 A. Yes. VGIC is confident that widespread EV exports can become a reality in response to
5 appropriate export credits for eligible customers, including along the anticipated
6 implementation timeline for SCE's VGRP (i.e., beginning in 2027). If the proposed
7 VGRP is adopted (with VGIC's recommended modifications) there are no fundamental
8 technical barriers that would prevent EVs from contributing. However, VGIC's optimism
9 in this regard is also tempered by the reality that V2G systems have not yet been
10 deployed at such a significant scale and that there are still some unknown factors
11 regarding overall customer acceptance and participation. We are encouraged by SCE's
12 2023 customer surveys, detailed in its Amended Rate Design Proposals, which provide
13 empirical evidence of customer acceptance and appetite for V2G energy exports,
14 detailing that 81% of surveyed customers expressed at least "a little interested." Notably,
15 unlike power plants that have a singular function, the primary role of EVs is to support
16 customers' transportation needs, not grid affordability, reliability, or decarbonization
17 needs. As such, to be successful the VGRP must be designed in a manner that puts the
18 customers' perspective first and foremost. VGIC believes this can and should be
19 achieved. Additionally, since this is a novel type of grid resource, any incremental
20 participation (even if small) will be beneficial and is still worth pursuing as part of a
21 comprehensive approach to advancing VGI pursuant to Senate Bill 676, which aims to
22 maximize feasible and cost-effective VGI by 2030.

Q. Is there a significant amount of EV and EV supply equipment (“EVSE”) available today that can already provide V2G capabilities?

A. Yes, although it is less widespread than V1G managed charging capabilities, a meaningful portion of EVs deployed today have bidirectional capabilities that would allow them to discharge. The most significant sources of V2G potential today are in the form of the light-duty passenger EVs and electric school buses. As of the end of 2024, the bidirectional EV and EVSE solutions detailed in *Table 1* below are available for customers and site developers. Beyond what is currently in existence, however, it is expected that several EV and EVSE models being launched prior to 2027 will have bidirectional charging capabilities. The table below summarizes both existing and forthcoming bidirectional vehicle models and charging equipment.

Table 1. Bidirectional-capable vehicles on California’s roads today, currently-deployed bidirectional EVSE, and forthcoming bidirectional products:

Product <i>* Denotes products that will soon be bidirectionally-capable</i>	Estimated Amount Deployed in California/ Anticipated Deployment Date	Notes
Nissan LEAF	55,349 total registered from 2013 through Q3 2024. ²	- MY 2013 or later are V2G-capable. - Bidirectional charging enabled with Fermata FE-20. ³

² California Energy Commission (2025). *New_ZEV_Sales_Last_update_11-19-2024_ada.xlsx*. ZEV and Infrastructure Stats Data. Data last updated September 30, 2024. Retrieved January 6, 2025 from <https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data>.

³ PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot*. <https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html>. Retrieved January 7, 2025.

Ford F-150 Lightning Electric	Estimated 22,296 total delivered through Q4 2024. ⁴	- Bidirectional charging enabled with Ford Charge Station Pro.
Chevrolet Silverado EV	Estimated 3,050 total delivered through Q4 2024. ⁵	- Bidirectional charging enabled with GM Energy PowerShift.
Chevrolet Equinox EV	Estimated 11,162 total delivered through Q4 2024. ⁶	- Bidirectional charging enabled with GM Energy PowerShift.
Chevrolet Blazer EV	Estimated 9,122 total delivered through Q4 2024. ⁷	- Bidirectional charging enabled with GM Energy PowerShift.
Cadillac Lyriq	Estimated 14,519 total delivered through Q4 2024. ⁸	- Bidirectional charging enabled with GM Energy PowerShift.
Tesla Cybertruck	6,670 total registered through Q3 2024. ⁹	- Bidirectional charging enabled with Tesla Universal Wall Connector.

⁴ Ford. *Ford U.S. Retail Sales Grow at Double the Industry Pace in 2024, Led by Trucks, Hybrids, Electric Vehicles and Lincoln*. January 3, 2025. https://s201.q4cdn.com/693218008/files/doc_news/2025/Jan/03/Ford-U-S-Q4-2024-Sales-Release.pdf. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

⁵ General Motors. *GM U.S. Deliveries for Quarter 4 2024*. January 3, 2025. <https://investor.gm.com/static-files/b4b3e0a7-f528-442a-89d4-443e62eae50b>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

⁶ *Ibid.*

⁷ *Ibid.*

⁸ *Ibid.*

⁹ California Energy Commission (2025). *New_ZEV_Sales_Last_update_11-19-2024_ada.xlsx*. ZEV and Infrastructure Stats Data. Data last updated September 30, 2024. Retrieved January 6, 2025 from <https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data>.

Kia EV9*	Estimated 8,944 total sold through Q4 2024. ¹⁰	- Bidirectional charging to be enabled soon with Wallbox Quasar 2. ¹¹
Volvo EX90*	Estimated 332 total sold through Q4 2024. ¹²	- Bidirectional charging to be enabled soon with dcbel Ara. ¹³
Lucid Motors Air	Estimated 6,279 total delivered through Q4 2024. ¹⁴	- Bidirectional charging enabled with RangeXchange charging adapter. ¹⁵
GMC Sierra EV*	Estimated 691 total delivered through Q4 2024. ¹⁶	- Bidirectional charging to be enabled soon with GM Energy PowerShift.

¹⁰ InsideEVs. *Kia Set New EV Sales Records Last Year in the U.S.* January 3, 2025.

<https://insideevs.com/news/746147/kia-ev-sales-record-2024-us/>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

¹¹ Kia EV9 x Wallbox Quasar 2. Retrieved January 7, 2025.

https://quasar2forkiaev9.wallbox.com/?utm_source=web&utm_medium=cta_button&utm_campaign=quasar2forkia

¹² Volvo Cars. *Sales Volumes: United States – Retail Sales by Car Model – December 2024*. Accessed January 7, 2025. <https://www.media.volvocars.com/us/en-us/corporate/sales-volumes>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

¹³ Electrek.co. *Volvo EX90 Bidirectional charging with DCBEL*. September 5, 2024.

<https://www.youtube.com/watch?v=wniV8rIObXU>

¹⁴ Lucid Motors. *Lucid Announces Fourth Quarter and Full Year 2023 Financial Results*. February 21, 2024.

<https://lucidmotors.com/media-room/lucid-q4-2023-financial-results> and Lucid Motors. *Lucid Announces Q4 Production & Deliveries, Sets Date for Fourth Quarter 2024 Results*. January 6, 2025.

<https://ir.lucidmotors.com/node/9736/pdf>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

¹⁵ Lucid Motors. *Introducing RangeXchange, an Innovative New Feature Enabling the Lucid Air to Directly Charge Other Electric Vehicles*. November 9, 2023. <https://lucidmotors.com/stories/introducing-rangexchange>

¹⁶ General Motors. *GM U.S. Deliveries for Quarter 4 2024*. January 3, 2025. <https://investor.gm.com/static-files/b4b3e0a7-f528-442a-89d4-443e62eae50b>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

Cadillac Escalade IQ EV*	Estimated 259 total delivered through Q4 2024. ¹⁷	- Bidirectional charging to be enabled soon with GM Energy PowerShift.
Cadillac Optiq*	Recently launched. Sales data not yet available.	- Bidirectional charging to be enabled soon with GM Energy PowerShift.
-	-	- Emporia V2X bidirectional charger coming soon. ¹⁸
BlueBird	428 buses through Q4 2023. ¹⁹	- 155 kWh or 196 kWh school buses ²⁰
Lion Electric	150 buses through Q4 2023. ²¹	- 126 kWh, 140 kWh, 168 kWh, or 210 kWh school buses ²²
BYD / RIDE	Estimated 102 buses through Q4 2024. ²³	- 156 kWh, 230 kWh, or 288 kWh school buses ²⁴

¹⁷ General Motors. *GM U.S. Deliveries for Quarter 4 2024*. January 3, 2025. <https://investor.gm.com/static-files/b4b3e0a7-f528-442a-89d4-443e62eae50b>. Assumes California comprises 38.66% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 7, 2025, listing 2,113,135 CA EV sales and 5,466,139 U.S. EV Sales. <https://www.veloz.org/ev-market-report/>

¹⁸ Emporia. *Coming Soon: V2X Bi-Directional Charger*. Accessed January 7, 2025. <https://www.emporiaenergy.com/how-the-emporia-v2x-charger-works/>

¹⁹ California Energy Commission (2024). Medium- and Heavy-Duty Zero-Emission Vehicles in California. Data last updated May 1, 2024. Retrieved January 7, 2025 from <https://www.energy.ca.gov/zevstats>

²⁰ BusinessWire. *Blue Bird Receives Record Order of 180 Electric School Buses*. January 16, 2024. <https://www.businesswire.com/news/home/20240116715215/en/Blue-Bird-Receives-Record-Order-of-180-Electric-School-Buses>

²¹ California Energy Commission (2024). Medium- and Heavy-Duty Zero-Emission Vehicles in California. Data last updated May 1, 2024. Retrieved January 7, 2025 from <https://www.energy.ca.gov/zevstats>

²² Lion Electric. *Vehicle-to-grid technology*. <https://schoolbus.thelionelectric.com/>. Accessed January 7, 2025.

²³ California Energy Commission (2024). Medium- and Heavy-Duty Zero-Emission Vehicles in California. Data last updated May 1, 2024. Retrieved January 7, 2025 from <https://www.energy.ca.gov/zevstats> and PG&E. *PG&E Helps Zum Deploy Nation's First 100% Electric School Bus Fleet in Oakland for New School Year*. August 27, 2024. <https://www.pge.com/en/newsroom/currents/future-of-energy/articles-4040-pge-helps-zum-deploy-nations-100-electric-school-bus-fleet-oakland-new-school-year.html>

²⁴ PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot*. <https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html>. Retrieved January 7, 2025.

Thomas Built	-	- 246 kWh school buses ²⁵
IC Bus	169 buses through Q4 2023. ²⁶	- 210 kWh or 315 kWh school buses ²⁷
Green Power Motors	-	- 118 kWh or 193 kWh school buses ²⁸
Phoenix Motorcars	-	- 100 kWh or 150 kWh school buses ²⁹
BorgWarner	-	- 60 or 125 kW V2G DC EVSE ³⁰
Tellus Power Green	-	- 20, 30, 40, or 60 kW V2G DC EVSE ³¹
InCharge	-	- 22, 44, or 66 kW V2G DC EVSE ³²
Heliox	-	- 60 kW V2G DC EVSE ³³

²⁵ Thomas Built Buses. *Electric school buses and utility companies: A powerful combination*. Retrieved January 7, 2025. <https://thomasbuiltbuses.com/electric-school-buses/utility-providers/>

²⁶ California Energy Commission (2024). Medium- and Heavy-Duty Zero-Emission Vehicles in California. Data last updated May 1, 2024. Retrieved January 7, 2025 from <https://www.energy.ca.gov/zevstats>

²⁷ IC Bus. *Electric CE Series School Bus*. https://www.icbus.com/-/media/Project/Navistar/ICBus/ICBus/electric_ce_series_brochure_spread_format.pdf. Retrieved January 7, 2025.

²⁸ California HVIP. *School Bus: Vehicle Type – ZESBI \$ Eligible*. <https://californiahvip.org/vehicle-category/school-bus/?type=655>. Retrieved January 7, 2025.

²⁹ *Ibid.*

³⁰ California Energy Commission. *Vehicle-to-Grid Equipment List (V2GEL)*. Retrieved January 7, 2025. <https://v2gel.energy.ca.gov/>

³¹ PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot*. <https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html>. Retrieved January 7, 2025.

³² InCharge. *Chargers & Hardware*. <https://inchargeus.com/chargers/>. Retrieved January 7, 2025.

³³ Heliox. *Heliox 60 kW*. <https://www.heliox-energy.com/us-products/60-kw-dc-charger>. Retrieved January 7, 2025.

1 **III. VGIC IS GENERALLY SUPPORTIVE OF THE VGRP CONCEPT, BUT**
2 **BELIEVES IT SHOULD BE IMPROVED UPON.**

3 **Q. Have you reviewed SCE’s VGRP proposal, and specifically, have you reviewed the**
4 **proposed rate design?**

5 A. Yes.

6 **Q. What is VGIC’s perspective on the VGRP proposal?**

7 A. VGIC is encouraged by this initial proposal and is interested in working with SCE and
8 other stakeholders to improve it and ultimately adopt it, with modifications.

9 **Q. Are there specific features of the proposal that you find especially attractive?**

10 A. Yes, there are several important features, including the following:

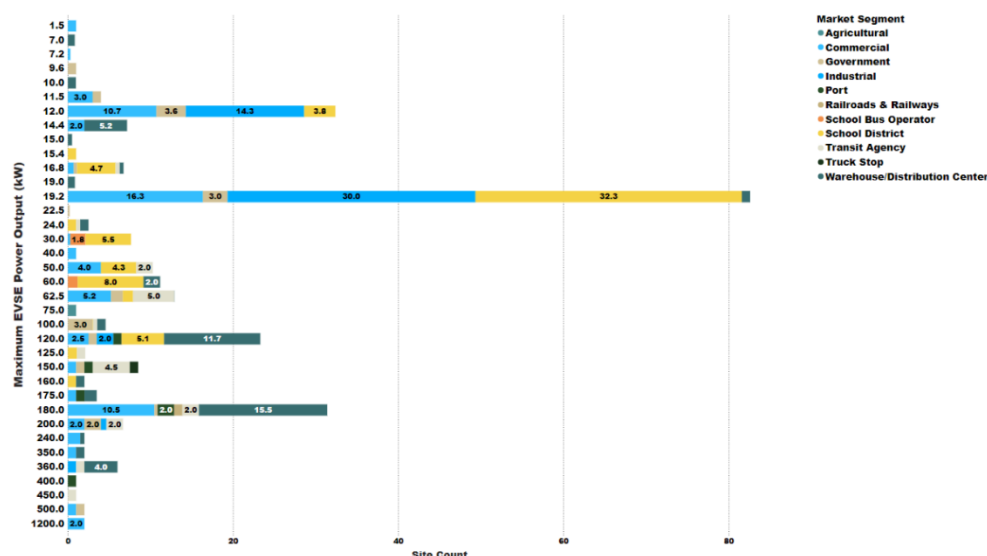
11 1) The establishment of a “stable and predictable export price by time-of-day.”³⁴ This has
12 the potential to support relatively widespread enrollment of customers with bidirectional
13 charging systems as pricing is based on pre-determined time-of-day pricing periods,
14 similar to the time-of-use (“TOU”) structure many customers are already familiar with.
15 The simplicity and stability of the TOU structure supports a strong customer
16 understanding for a range of participants, from early adopters to mass-market
17 participants. Furthermore, the TOU export credit would be made available during peak
18 hours throughout the summer, which provides customers with an ample opportunity to
19 support the grid through exports from bidirectional charging systems.

³⁴ SCE-04A at 129.

2) The broad applicability to residential and non-residential customers casts an appropriately wide net for interested customers. This is appropriate because the ways in which EV drivers use their vehicles are diverse, even within certain customer classes or vehicle segments. For example, the below figure shows the diverse EVSE power output and market segments of customers in SCE’s Charge Ready Transport program:

Figure 1: Southern California Edison Charge Ready Transport EVSE as of November 2024³⁵

EVSE Power Output Levels by Market Segment



Note, figures reflect active, committed sites with signed CRT Program Agreements. Partial site counts represent sites with multiple types of charging equipment.

As an additional example, the below figure shows the various “dimensions” considered in the 2019-2020 VGI Working Group *Use Case Assessment*. The effort convened about 100 representatives from utilities, state and federal agencies, California Independent

³⁵ Southern California Edison. *Transportation Electrification Program Advisory Council*. December 13, 2024. Page 14.

System Operator, automakers, charging manufacturers, charge management system providers, charge point operators, transit agencies, research institutions, industry associations, environmental nonprofits, consumer groups, ratepayer advocates, and standards development organizations to define, assess value, and rank 2,500 potential VGI use cases.

Figure 2: 2019-2020 VGI Working Group Report Use Case Assessment Framework³⁶

TABLE 2. Dimensions of the Use Case Assessment Framework and Use-Case-Definition Options

SECTOR	APPLICATION	TYPE	APPROACH	RESOURCE ALIGNMENT
Residential-Single-Family Home	Customer-Bill Management	V1G	Indirect (passive)	Unified and Aligned
Residential-Single-Family Home, Rideshare	Customer-Upgrade Deferral	V2G	Direct (active)	Fragmented and Aligned
Residential-Multi-Unit Dwelling	Customer-Backup, Resiliency			Fragmented and Misaligned
Residential-Multi-Unit Dwelling Rideshare	Customer-Renewable Self-Consumption			
Commercial-Workplace	System-Grid Upgrade Deferral			
Commercial-Public, Destination	System-Backup, Resiliency			
Commercial-Public, Destination Rideshare	System-Voltage Support			
Commercial-Public, Commute	System-Day-Ahead Energy			
Commercial-Public, Commute Rideshare	System-Real-Time Energy			
Commercial-Fleet, Transit Bus	System-Renewable Integration			
Commercial-Fleet, School Bus	System-GHG Reduction			
Commercial-Fleet, Small Truck (class 3-5)	System-RA, System Capacity			
Commercial-Fleet, Large Truck (class 6-8)	System-RA, Flex Capacity			
	System-RA, Local Capacity			
	System-Frequency Regulation Up/Down			
	System-Spinning Reserve			
	System-Non-Spinning Reserve			

Given the *considerable* diversity of relevant use cases, offering an open, optional, and predictable export rate rider to a wide set of customers allows them to participate in the way that best fits their specific needs and internal cost-benefit calculations.

³⁶ See Final Report of the VGI Working Group. <https://gridworks.org/2020/09/final-report-of-the-california-joint-agencies-vehicle-grid-integration-vgi-working-group/>

1 3) The cost-based, long-term nature of the proposed VGRP can underpin a degree of
2 certainty for bidirectional charging supply chain development, distribution channels,
3 customer engagement, and overall ecosystem that has never been achieved through the
4 few one-off pilot projects that have been authorized to date.

5 4) The eligibility of customers with and without NEM/NBT technologies supports the
6 diverse set of customers that may be interested in enrolling in VGRP, while also
7 recognizing the observed correlation between EV and rooftop solar PV owners.³⁷

8 **Q. Are all of these features found in any other opportunities for bidirectional charging**
9 **customers in SCE’s service territory?**

10 A. No. There are four alternative options for SCE customers with bidirectional charging
11 equipment:

12 (1) SCE Emergency Load Reduction Program. While ELRP offers a predictable
13 \$2/kWh credit, the event-based nature of the program is unpredictable relative to
14 the pre-determined peak periods proposed in VGRP. Additionally, ELRP is not a
15 sustainable, cost-based measure. It is currently authorized through 2027.

16 (2) California Energy Commission (“CEC”) Demand Side Grid Support (“DSGS”) –
17 Option 3. This program allows aggregators to enroll customers in a summer
18 reliability program in exchange for capacity payments, which are awarded when
19 the aggregators dispatch customer resources via test events or in response to
20 certain event triggers. Similar to ELRP, the DSGS Option 3 program is not a
21 sustainable, cost-based offering, and is only available until funding, determined
22 by the legislature and governor, is no longer available. Funding is shared across
23 all options and resources in DSGS.

³⁷ Sharda, Garikapati, et al. *The electric vehicles-solar photovoltaics Nexus: Driving cross-sectoral adoption of sustainable technologies*. Renewable and Sustainable Energy Reviews, Volume 191, March 2024.
<https://www.sciencedirect.com/science/article/pii/S1364032123010304?dgcid=author>

1 (3) Direct CAISO participation via third-party providers. Behind-the-meter energy
2 storage systems have been generally slow to participate in the CAISO market due
3 to a number of barriers, including the complexity and cost of doing so. VGIC is
4 unaware of any bidirectional charging sites across any customer class in
5 California currently participating directly in CAISO.

6 (4) SCE Dynamic Rate Pilots. As detailed in SCE’s Additional Rate Design
7 Proposals and Cal Advocate’s Prepared Testimony of Additional Rate Design
8 Proposals, SCE offers dynamic rate options to certain customers. These dynamic
9 rate options do not offer stability or predictability due not only to the hourly, day-
10 ahead nature of the rate design, but also the utilization of a “subscription” or
11 baseline that complicates the customer’s ability to estimate its savings due to
12 participating in the dynamic rate option. Moreover, SCE currently has no
13 bidirectional charging systems enrolled in either the first or second phases of
14 SCE’s dynamic rate pilots.³⁸

15 **Q. Is there data demonstrating that SCE V2G customers are particularly interested in**
16 **the current iteration of SCE’s dynamic rate pilot?**

17 A. VGIC is not aware of any data demonstrating V2G customer interest in SCE’s dynamic
18 rate pilot. SCE notes that there were no bidirectional charging systems enrolled in the
19 first phase of SCE’s Dynamic Rate Pilot.³⁹ Based on conversations with VGIC’s 37
20 members, we understand there to be concerns over the current implementation of SCE’s
21 dynamic rate pilot, including the lack of automated service providers with exclusive
22 focus on integrating bidirectional charging customers, the “subscription” or baseline
23 element’s muting effect on the dynamic price signal, and the complexity of providing

³⁸ Attachment A. VGIC-SCE-001. Response to Question 03.a-c.

³⁹ Attachment A. VGIC-SCE-001. Response to Question 03.a-c.

1 customers with an accurate estimate of their expected savings under the dynamic rate
2 pilot.

3 **Q. Please explain the concept of path dependence and how it relates to V2G rate design**
4 **decisions made today.**

5 A. “Path dependence” refers to the influence of past decisions on future outcomes, hindering
6 the implementation of alternative solutions. In the case of V2G customers, VGIC is
7 concerned that relying too heavily on a single dynamic rate approach for all customers, a
8 recommendation made in Cal Advocates’ testimony and detailed further in Section V
9 below, would lock in a specific future that may not, in fact, be in the best interest of
10 ratepayers. Notably, Cal Advocates has published findings suggesting that optimized VGI
11 implementation can reduce electrification-driven distribution system costs from the
12 Commission’s estimated \$50 billion through 2035 down to \$22.7 billion.⁴⁰ Given the
13 magnitude of these numbers, the stakes are incredibly high, especially for low-income
14 and other vulnerable California ratepayers already feeling an outsized energy cost burden.
15 However, to achieve the reductions modeled by Cal Advocates, IOUs will need to engage
16 EV customers at a significant enough scale for these optimized load shapes to have a
17 beneficial impact on the total required distribution system upgrades. VGIC is concerned
18 that pursuing only dynamic rates, with the only available evidence showing that V2G
19 customers are not interested in participating,⁴¹ risks a gamble with path dependence,
20 wherein SCE would be “locked in” to offering only dynamic rates, with no opportunity to

⁴⁰ See CPUC Electrification Impacts Study Part 1:

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M508/K423/508423247.PDF> and Cal Advocates DGEM 2.0 Preliminary Results, October 21, 2024. <https://www.publicadvocates.cpuc.ca.gov/-/media/cal-advocates-website/files/press-room/reports-and-analyses/241024-public-advocates-office-dgem-20-preliminary-results.pdf>

⁴¹ Attachment A. VGIC-SCE-001. Response to Question 03.a-c.

1 understand V2G customer interest in relatively simple, predictable static export credits
2 until it is too late to avoid the upgrades forecasted by 2035 and the resulting ratepayer
3 costs of up to \$50 billion.

4
5 **IV. RECOMMENDED MODIFICATIONS TO THE PROPOSED VGRP TO**
6 **ADDRESS DEFICIENCIES IN SCE’S PROPOSAL AND CONCERNS RAISED**
7 **BY CAL ADVOCATES.**

8
9 **Q. Are there features of the proposed VGRP you believe are insufficient?**

10 **A.** Yes. There are two main elements of VGRP that are insufficient. One element is detailed
11 herein, and the other is described further below in this section (i.e., Section IV).

12 The proposal arbitrarily limits the applicable avoided cost calculator (“ACC”) values
13 used to calculate VGRP compensation to the following five components: (1) GHG Cap
14 and Trade (Marginal Emissions); (2) Generation Energy; (3) Generation Capacity; (4)
15 Ancillary Services Procurement; and (5) Losses. SCE notes “because SCE’s proposal is
16 intended as a first step on the path to enabling this still nascent market segment, the
17 following ACC components are not included in SCE’s V2G export price: (1) GHG Adder
18 (Marginal Emissions); (2) GHG Portfolio Rebalancing; (3) Transmission; (4)
19 Distribution; and (5) Methane Leakage.”⁴² SCE offers no substantive justification in its
20 testimony for why it has chosen to include certain ACC components but not others. In

⁴² SCE-04A at 132.

1 response to VGIC’s data request, SCE explains that transmission and distribution
2 components were not included because of the “inability to incorporate EV exports into
3 the planning of the grid and T&D system.” SCE asserts that V2G resources are distinct
4 from stationary behind-the-meter (“BTM”) resources, which SCE can reliably predict.⁴³

5 **Q. What is the purpose of the CPUC’s adopted ACC?**

6 A. As we understand it, the purpose of the ACC is to provide an estimate of the value of a
7 kWh delivered to the grid.

8 **Q. Does VGIC agree with SCE’s position that kWh generated from V2G systems yield**
9 **less avoided transmission and distribution costs than kWh from other BTM**
10 **resources?**

11 A. No. VGIC does not agree with this position and finds that differences in predictability of
12 customer behavior are not a reasonable rationale to exclude V2G customers from certain
13 ACC components. VGIC finds it unreasonable that these ACC components should be
14 excluded from the proposed VGRP value stack but included in export compensation for
15 over 40,000 other customers receiving export compensation based on ACC in SCE’s
16 territory.⁴⁴

17 **Q. Are you aware of any instances where SCE offers export compensation containing**
18 **both generation and distribution components to bidirectional charging customers?**

19 A. Yes. SCE’s Dynamic Rate Pilot eligibility is open to bidirectional charging customers
20 and its rate design compensates exports under a rate that includes a distribution

⁴³ Attachment A. VGIC-SCE-001. Response to Question 02.a-e.

⁴⁴ Attachment A. VGIC-SCE-001. Response to Question 02.a-e.

1 component.⁴⁵ Similarly, V2G exports in Pacific Gas & Electric (“PG&E”) Dynamic Rate
2 Pilot Phase 2⁴⁶ and V2X Pilots Phase 2⁴⁷ will be compensated under an export rate that
3 includes a distribution component. SCE does not offer an explanation as to why V2G
4 exports in the Dynamic Rate Pilot are offered both generation and distribution
5 components but V2G exports in the proposed VGRP would not be.

6 **Q. Do you know of any reason why the full stack of avoided costs contained in the ACC**
7 **would differ based on the type of resource used to generate the electrons that are**
8 **then used on the grid?**

9 A. No. The resource from which the kWh was generated has no bearing on the value of
10 avoided costs resulting from that kWh. For either customers eligible for the Net Billing
11 Tariff (“NBT”) or customers eligible for VGRP, the totality of ACC components should
12 be included in determining the value and compensation of the exported kWh.

13 **Q. Please detail any other deficiencies found in VGRP.**

14 A. As proposed by SCE, VGRP customers with Net Energy Metering (“NEM”) or NBT
15 eligible systems will need to install a V2G Net Generation Output Meter (“V2G-
16 NGOM”) to separate out V2G exports from NEM/NBT-eligible system exports and apply
17 appropriate pricing to each. SCE also proposes to install V2G-NGOMs for customers
18 without NEM or NBT eligible systems to “prepare the customer for any future NBT

⁴⁵ See, for example, SCE Advice Letter 4684-E *SCE’s Dynamic Rate Pilot Pursuant to D.21-12-015*. January 5, 2022. <https://www.dret-ca.com/wp-content/uploads/2022/02/Advice-Letter-4684-E.pdf> and D.24-01-032 *To Expand System Reliability Pilots of PG&E and SCE*. January 26, 2024. <https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M524/K176/524176497.PDF>. Pg. 36.

⁴⁶ *Id.* at 8-36.

⁴⁷ PG&E Advice Letter 6694-E. *Rate Structured for Vehicle Grid Integration Pilots*. September 2, 2022. https://www.pge.com/tariffs/assets/pdf/adviceletter/ELEC_6694-E.pdf

installations.”⁴⁸ VGIC finds two deficiencies here. First, SCE’s justification for installing V2G-NGOM at customer sites who do not have NEM/NBT-eligible systems is unnecessary. SCE fails to provide adequate justification for requiring installation of a V2G-NGOM at a customer site solely because that customer may one day have a NEM/NBT-eligible solution.

Second, requiring V2G-NGOM at all sites, including both NEM/NBT and non-NEM/NBT sites, adds significant cost (i.e., over \$1,100)⁴⁹ and potential delays. These costs can especially undermine the project economics of smaller V2G sites, which have relatively low upfront capital expenditures. The negative customer impacts of the NGOM requirement could be mostly avoided by adopting VGIC’s proposed modifications, detailed below.

Q. What are VGIC’s proposed modifications to the VGRP proposal, and how does it address the deficiencies found in SCE’s VGRP?

A. VGIC’s proposed modifications to the VGRP proposal contains three key modifications to SCE’s VGRP, detailed below:

(1) VGIC proposes VGRP be modified to incorporate all ten ACC components, rather than the list of five arbitrary ACC components proposed by SCE.

(2) VGIC proposes VGRP price signals mirror NBT price signals. More detail and justification are provided below:

(a) Based on VGIC’s understanding, the illustrative rates detailed in SCE’s Table IX-31 were achieved by taking the five proposed ACC values grouped by

⁴⁸ SCE-04A at 130-131.

⁴⁹ SCE Unit Cost Guide dated March 31, 2022. https://www.sce.com/sites/default/files/custom-files/Web%20files/Attachment_A-Unit_Cost_Guide.pdf

1 hour and month to create averages in the ACC values for the proposed peak
2 time periods. The resulting eight time periods (summer weekday off-peak vs
3 on-peak, summer weekend off-peak vs on-peak, winter weekday off-peak vs
4 on-peak, and winter weekend off-peak vs on-peak) reflect these average ACC
5 values.

6 (b) Instead of this methodology, VGIC recommends VGRP customers be
7 compensated for exports pursuant to the NBT design (or any NBT
8 successor(s) should they be approved). This builds on and is consistent with
9 proposed modification 1 above: to incorporate all ten ACC components, rather
10 than the five arbitrary components proposed by SCE.

11 (c) Critically, this would largely obviate the need for NGOMs since all exports –
12 whether from EVs or NBT-eligible devices – would be compensated at the
13 same rate thus eliminating the need to determine which resource a particular
14 kWh of export came from.

15 (d) This would also simplify billing system upgrades, the cost of which is one of
16 the primary concerns raised by Cal Advocates (and discussed below in
17 Section V) in its testimony recommending SCE's proposed VGRP be rejected.

18 (e) As detailed by Cal Advocates (and discussed below in Section V), VGRP as
19 proposed is less granular than other opportunities and, based on Cal
20 Advocates' assessment, does not sufficiently reflect accurate grid conditions.
21 By modifying the proposal to incorporate 576 ACC-based values annually,
22 rather than 8 ACC-based values, the VGRP will much more closely reflect
23 grid conditions.

24 (3) VGIC proposes that NEM 1 and NEM 2 customers be eligible for VGRP only if
25 they: (i) install a NGOM, (ii) switch to NBT, (iii) install a device approved
26 pursuant to the Commission's *PEV Submetering Protocol*, or (iv) elect another
27 solution developed bilaterally between SCE and relevant OEMs or otherwise
28 made optionally available by SCE.

1 (a) This proposal may motivate certain legacy NEM 1 and NEM 2 customers to
2 switch to a more dynamic rate with greater grid benefits (i.e., NBT), as a way
3 to avoid the costs of installing a NGOM (option (i)) or metering alternatives
4 (options (iii) or (iv)).

5 (b) The *PEV Submetering Protocol*, adopted by the CPUC in August 2022, allows
6 for embedded bidirectional EVSE submeters or external bidirectional EVSE
7 submeters to be used upon customer request, assuming the requested device is
8 an approved non-utility-owned submeter and that a relevant meter data
9 management agent (“MDMA”) has also been approved by SCE. Embedded or
10 external submeters can therefore provide a low-cost alternative to a V2G-
11 NGOM. Notably, this device-level metering is already implemented in
12 California through the CEC’s DSGS Option 3, as well as in New England
13 under National Grid and Eversource’s Connected Solutions, Daily Dispatch
14 programs.

15 (c) Regarding options (i) and (iii), VGIC notes that “SCE has not yet deployed
16 any NGOMs at customer bidirectional charging sites in SCE service
17 territory.”⁵⁰ However, SCE has approved 28 EVSE embedded submeters and
18 2 non-embedded submeters, and has recently approved the first Meter Data
19 Management Agent (“MDMA”) per the *PEV Submetering Protocol*.⁵¹

20 **Q. Does VGIC believe that, absent these changes, the proposed VGRP compensation**
21 **level is sufficient to support meaningful participation?**

22 A. No. VGIC believes that SCE’s proposed VGRP, absent VGIC’s recommended changes,
23 would not sufficiently attract meaningful customer participation. The Annual Customer
24 Estimated Credits table provided by SCE⁵² presents the potential for customer
25 remuneration for V2G exports. However, based on VGIC’s discussions with its 37

⁵⁰ Attachment A. VGIC-SCE-001. Response to Question 01.a.

⁵¹ Attachment A. VGIC-SCE-001. Response to Question 01.b-e.

⁵² SCE-04A at 134.

1 members, including automakers, V2G aggregators, and EVSE manufacturers, it is
2 unlikely that the values are high enough to attract meaningful participation. Furthermore,
3 the values may not be high enough to cover the upfront costs associated with deploying
4 the necessary V2G equipment to participate in the rate. Especially as the V2G market is
5 relatively nascent, VGIC is concerned that the estimated credits listed in SCE's Table IX-
6 32, i.e. credits comprising only five of the ten total ACC components, are not sufficient to
7 overcome the upfront cost barriers of V2G sites, which often include special chargers,
8 bidirectional inverters, communications devices, customer panel upgrades, and related
9 electrical components and installation costs.

10 VGIC has witnessed several utilities make initial forays into proposing and developing
11 compensation mechanisms for VGI solutions, such as compensation for V2G exports.

12 However, in some cases these attempts appear to be simply unworkable for EV customers
13 and may in fact yield a net cost for V2G behavior (rather than a net benefit). While
14 experimentation with new compensation mechanisms should be encouraged, VGIC is
15 concerned that if inadequate mechanisms are adopted, they could significantly delay the
16 development of other mechanisms that are more fully considered and could meaningfully
17 spur the VGI market. As a result, improvements should be made to cure the deficiencies
18 identified above before adopting VGRP.

19
20 **V. ADDRESSING ISSUES RAISED BY CAL ADVOCATES REGARDING VGRP**
21 **IMPLEMENTATION.**

1 **Q. In several responses above you referenced Cal Advocates’ positions on VGRP. Can**
2 **you describe your understanding of Cal Advocates positions regarding VGRP?**

3 A. Yes. Cal Advocates’ *Prepared Testimony of Additional Rate Design Proposals in SCE’s*
4 *2025 GRC Phase 2* contains two key recommendations related to SCE’s VGRP. The first
5 is to reject SCE’s VGRP, as SCE has not provided an adequate budget proposal. The
6 second is to “require SCE to expand the list of eligible rates in the Expanded Pilot
7 authorized by Decision (D.) 24-01-034 to include those that SCE proposed for
8 participation in its VGRP.”⁵³

9 **Q. Cal Advocates asserts that SCE does not provide budget estimates for each**
10 **identified budget category, and that this justifies rejecting SCE’s proposed VGRP.**
11 **Does VGIC agree with this assertion?**

12 A. Yes and no. While it is correct that SCE has not provided budget estimates for each
13 identified budget category, VGIC does not agree that this is an adequate reason to reject
14 SCE’s proposed VGRP. It is not unprecedented to refine budget details throughout the
15 course of a litigated proceeding, for utilities or program administrators to provide budgets
16 after program approval in subsequent implementation advice letters, or for utilities to be
17 authorized to record costs in memorandum accounts that are later evaluated by the
18 Commission and stakeholders in a subsequent cost recovery proceeding. For example,
19 PG&E’s DAHRTP export rate pilot budget was first proposed in the settlement, rather
20 than the utility’s testimony or application.⁵⁴ Another example is found in budget requests

⁵³ Cal Advocates. *Prepared Testimony of Additional Rate Design Proposals in SCE’s 2025 GRC Phase 2*. 9-1.

⁵⁴ *Joint Motion of the Public Advocates Office at the CPUC, VGIC, Electrify America, and PG&E for Adoption of Joint Settlement Agreement in PG&E’s Day Ahead Real Time Pricing Commercial EV Proceeding on Non-NEM Export Compensation Pilot*. June 17, 2022.

<http://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M488/K538/488538216.PDF>. Page 20.

1 for the California Energy Commission’s Gas Research and Development Program, which
2 are made via Tier 3 Advice Letter.⁵⁵ For community solar programs (Disadvantaged
3 Communities Green Tariff and Community Solar Green Tariff Programs), a broad budget
4 cap was adopted but specific annual budgets for program administration are submitted
5 annually via Tier 2 Advice Letter.⁵⁶ Costs for implementation of the Net Energy Metering
6 and Net Billing Tariffs are recorded in memorandum accounts that are later included in
7 subsequent GRCs for cost recovery.⁵⁷

8 Furthermore, if VGIC’s proposed modifications to VGRP, detailed in Section IV above,
9 are adopted, it is worth noting that implementation costs will likely be lower, as VGRP
10 customers would receive price signals that mirror NBT (or any NBT successor), which
11 likely mitigates billing system upgrade costs. Offering the same NBT prices to VGRP
12 customers may also present opportunities to mitigate certain marketing, education, and
13 outreach costs, as existing NBT customers could be expected to be familiar with the NBT
14 price approach.

15 **Q. Cal Advocates states “SCE refers to PG&E’s day-ahead hourly real-time pilot rate**
16 **(DAHRTP) as reason to approve SCE’s VGRP. However, PG&E’s application for**
17 **its DAHRTP had a cost forecast and implementation plan for its pilot rate.”⁵⁸ Cal**
18 **Advocates cites PG&E’s application filed October 23, 2020. Cal Advocates further**
19 **asserts that “PG&E’s DAHRTP implementation has been delayed due to its overlap**

⁵⁵ CPUC Resolution G-3571 issued November 5, 2020.

<https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M347/K955/347955274.PDF>

⁵⁶ CPUC Resolution E-5125 issued August 6, 2021.

<https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M398/K239/398239873.PDF>

⁵⁷ D.22-12-056 at Ordering Paragraph 12.

<https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M500/K043/500043682.PDF>

⁵⁸ Cal Advocates. *Prepared Testimony of Additional Rate Design Proposals in SCE’s 2025 GRC Phase 2*. 9-11.

1 **with the Expanded [Dynamic Rate] Pilots.”⁵⁹ Does Cal Advocates correctly**
2 **represent the order of events in A.20-10-011 and the reason for delaying DAH RTP**
3 **implementation?**

4 A. Cal Advocates misrepresents the order of events in the DAH RTP proceeding and the
5 reason for delaying DAH RTP implementation.

6 First, Cal Advocates conflates the development of the DAH RTP application, a dynamic
7 *charging* rate for commercial EV customers, with the subsequent development of the
8 DAH RTP *export* rate pilot, which came later and is the item most relevant to SCE’s
9 VGRP proposal. Whereas VGRP is focused on export credits for EV customers from the
10 outset, PG&E’s original DAH RTP application did not contain *any* pilot or rate proposal
11 for export credits for EV customers. All pilot design and implementation details,
12 including budget estimates, for the DAH RTP *export pilot* emerged approximately two
13 years later following two party settlements and subsequent Commission decisions in that
14 proceeding.⁶⁰ As such, VGIC reiterates that the Commission should disregard Cal
15 Advocates’ argument that SCE’s VGRP proposal should be rejected due to a lack of
16 estimated budget in SCE’s initial proposal.

17 Second, Cal Advocates’ claim that “PG&E’s DAH RTP implementation has been delayed
18 due to its overlap with the Expanded Pilots” is incorrect. PG&E’s DAH RTP CEV export
19 pilot has been unable to launch pilot due to delays in billing system upgrades, as
20 explained by PG&E: “PG&E was unable to meet these target launch dates due to their

⁵⁹ Cal Advocates. *Prepared Testimony of Additional Rate Design Proposals in SCE’s 2025 GRC Phase 2*. 9-11.

⁶⁰ *Joint Motion of the Public Advocates Office at the CPUC, VGIC, Electrify America, and PG&E for Adoption of Joint Settlement Agreement in PG&E’s Day Ahead Real Time Pricing Commercial EV Proceeding on Non-NEM Export Compensation Pilot*. June 17, 2022.

<http://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M488/K538/488538216.PDF>. Page 20.

1 dependence on the deployment of PG&E’s multi-year billing system upgrade.”⁶¹ VGIC
2 believes this further supports the modification to VGRP detailed in Section IV above to
3 mirror NBT prices, as this would likely mitigate the risk of SCE experiencing similar
4 billing system upgrade delays that PG&E is experiencing as it attempts to implement an
5 entirely new rate design.

6 **Q. Cal Advocates states SCE should be required to include rate schedules proposed to**
7 **be included in VGRP in the Expanded Pilots instead of implementing VGRP, as this**
8 **will allow for more participation in the Expanded Pilots. Do you agree?**

9 A. No, I disagree. While more rate eligibility in the Expanded Pilots may seem to support
10 more V2G customer participation, this is not necessarily the case. In fact, the only
11 available evidence related to this topic shows zero V2G customers enrolled in SCE’s
12 dynamic rate pilot.⁶² VGIC offers two potential reasons for the absence of enrolled V2G
13 customers in SCE’s dynamic rate pilot.

14 First, the complexity of the Expanded Pilots and the unpredictable customer savings
15 participation may yield, combined with the complexity of V2G technology, which is new
16 to nearly all customers, may create a compound barrier that could be otherwise avoided
17 by offering a simpler, more predictable rate design alternative, such as VGRP.

18 Second, customers will likely partner with OEMs or other service providers to facilitate
19 participation in either VGRP or Expanded Pilots. The Expanded Pilots are designed
20 around automated service providers (“ASP”) that win an SCE Request For Proposals

⁶¹ *Response of PG&E to Provide Information Required in Executive Director Letter Dated February 28, 2024.* August 28, 2024. <http://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M539/K203/539203335.PDF>. Attachment - Page 1.

⁶² Attachment A. VGIC-SCE-001. Response to Question 03.a-c.

1 (“RFP”). Notably, the terms of the RFP issued in September 2024 favored larger,
2 incumbent ASPs (e.g., smart thermostat aggregators), rather than newer, smaller V2G
3 ASPs. Meanwhile, VGRP customers could optimize charging and discharging using
4 whichever OEM/ASP/third-party partner of their choosing, providing more optionality
5 for customers and, in turn, supporting more widespread customer participation.

6 **Q. Cal Advocates asserts that enhancing participation in the Expanded Pilot will**
7 **provide a more dynamic price signal than VGRP, and that export credits in the**
8 **Expanded Pilot are “more closely aligned with the needs of the grid since they**
9 **reflect the real-time value of energy...Expanded Pilot would provide customers with**
10 **a more accurate price signal...”⁶³ Do you agree that the Expanded Pilots yield more**
11 **accurate price signals?**

12 A. No. While the CalFUSE framework underpinning the Expanded Pilots is designed to
13 offer more accurate price signals, VGIC notes that ACC, although based on forecasted
14 values, leverages a transparent ACC tool for which both the values and methodology
15 have been litigated by parties for several years and rules on by the Commission. In
16 contrast, the Expanded Pilot price signals are computed by a third-party vendor under
17 contract directly with the utility, with no published information or stakeholder
18 engagement behind the inputs or methodology used to arrive at hourly values. The
19 Expanded Pilots offers customers and/or ASPs a final day-ahead value for each hour, but
20 offers less transparency, accountability, and stakeholder engagement opportunity than the
21 ACC tool. While VGIC hopes the Expanded Pilot participants in both SCE and PG&E
22 service territory are receiving accurate prices, the current mechanism for calculating these

⁶³ Cal Advocates. *Prepared Testimony of Additional Rate Design Proposals in SCE’s 2025 GRC Phase 2*. 9-12.

prices is opaque and offers no pathway for stakeholders to assess the accuracy of the price signals.

Q. Do you believe that further expanding SCE’s Expanded Pilot would yield optimal V2G participation and the resulting grid benefits?

A. No. As detailed in Section III above, the ways in which EV drivers use their vehicles are diverse, and the ways in which these vehicles interact with the grid are perhaps even more diverse. To transition the VGI market from early V2G demonstrations and pilots to large-scale deployments that can meaningfully support the grid and provide grid edge energy storage capacity at scale, V2G customers may benefit from access to a *menu* of export compensation options. EV customers have a diverse set of needs and serve a variety of use cases characterized by various duty cycles, dwell times, vehicle battery sizes, charger configurations, and site load characteristics. As noted above in Section III, the 2019-2020 VGI Working Group defined, sorted, assessed, and ranked 2,500 potential VGI use cases. Meanwhile, SCE’s Charge Ready Transport program, the nation’s largest medium- and heavy-duty make-ready charging program, demonstrates a rich array of various customer segments and charger sizes.

Promoting grid-friendly behavior, including V2G exports, from an exceptionally diverse subset of utility customers presents a unique challenge. This use case diversity strongly suggests that no one single compensation mechanism will optimally capture all potential beneficial V2G participation. This is consistent with the argument made by SCE in its VGRP proposal:⁶⁴ holistic frameworks and strategies consisting of a mix of static retail

⁶⁴ SCE-04A at 126-130.

1 rate options, dynamic rate pilots, even-based demand response programs, direct control
2 programs, and wholesale market participation allows customers the choice to participate
3 in the option that best fits their specific needs and internal cost-benefit calculations. For
4 some customers, demand response strategies and relatively infrequent event-based
5 programs may be better suited to their needs, goals, and technologies. Other customers
6 may be attracted to a 24/7, year-round optimization approach. Yet other customers may
7 seek to dual participate in both utility export rates and non-rate utility programs, where
8 permitted, to maximize cost-effectiveness. In this way, there appears to be no one-size-
9 fits-all approach to unlocking V2G capability to support the grid. Just as grid planners
10 seek fuel diversity in the bulk power generation profile, the Commission should seek
11 V2G strategy diversification. This is especially critical within the context of the relatively
12 nascent V2G market, where there is a need to prompt initial customer response from a
13 wide array of new vehicle classes and use cases, customer segments, charger sizes and
14 types, site hosts, relevant agents and actors, and price sensitivities.

15 Furthermore, by offering a portfolio of participation options, the Commission and utilities
16 can experiment with innovative rate design approaches, like dynamic rate pilots, while
17 still maintaining the key rate design principles of gradualism, simplicity,
18 understandability, public acceptability, and feasibility of application through offering
19 widely used export compensation mechanisms. Critically, pursuing these options in
20 parallel mitigates against path dependence, as detailed above in Section III.

21 **Q. Based on the above responses, do you agree with Cal Advocates' recommendation to**
22 **add rate schedule eligibility into the Expanded Rate pilots in lieu of adopting VGRP**
23 **in this proceeding?**

1 A. No, I do not agree with Cal Advocates' proposal to add rate schedule eligibility into the
2 Expanded Rate pilot in lieu of adopting SCE's VGRP, as modified by VGIC's
3 recommendations in Section IV above.

4 **Q. Are there any other reasons why you disagree with Cal Advocates' recommendation**
5 **to add additional rate schedule eligibility into the Expanded Rate pilots in lieu of**
6 **adopting VGRP?**

7 A. Yes. SCE's Expanded Pilot is approved by D.24-01-032, and SCE recently filed a
8 separate rate application to address further development of its dynamic rates. As such, it
9 is not reasonable to impose additional requirements on the in-development Expanded
10 Pilot or establish requirements for new dynamic rates that are being litigated in SCE's
11 separate application.

12

13 **VI. CONCLUSION**

14 **Q. Does this conclude your testimony?**

15 A. Yes.

Appendix A:

Data Request VGIC-SCE-001

Southern California Edison
A.24-03-019 – 2025 SCE GRC Phase 2

DATA REQUEST SET V G I C - S C E - 0 0 1

To: VGIC-Vehicle-Grid Integration Council
Prepared by: Blake Evan Heidenreich
Job Title: Advisor
Received Date: 12/27/2024

Response Date: 1/7/2025

Question 01.a:

In SCE-04A, SCE notes a V2G Net Generation Output Meter (V2G-NGOM) will determine EV exports separately from NEM or NBT exports. SCE also states a V2G-NGOM will be used to measure EV exports for customers without NEM or NBT eligible systems.

a. Please detail the quantity of NGOM SCE has deployed at bidirectional charging customer sites in its service territory.

Response to Question 01.a:

- a. SCE has not yet deployed any NGOMs at customer bidirectional charging sites in SCE service territory. There are no known EV owners in SCE's service area today seeking compensation from exporting their EV's energy to the grid as there is no compensation mechanism for this behavior. There are customers in SCE territory installing bidirectionally capable systems that currently are only non-export (non-V2G). These customers may be interested in V2G exporting if appropriate tariffs and rules allowing compensation were available for customer enrollment. It is SCE's understanding that non-export bidirectional systems may only require a software update and not additional hardware to be capable of exporting energy to the grid.

Southern California Edison
A.24-03-019 – 2025 SCE GRC Phase 2

DATA REQUEST SET V G I C - S C E - 0 0 1

To: VGIC-Vehicle-Grid Integration Council
Prepared by: Paul Kasick
Job Title: Sr. Project Manager
Received Date: 12/27/2024

Response Date: 1/7/2025

Question 01.b-e:

In SCE-04A, SCE notes a V2G Net Generation Output Meter (V2G-NGOM) will determine EV exports separately from NEM or NBT exports. SCE also states a V2G-NGOM will be used to measure EV exports for customers without NEM or NBT eligible systems.¹

b. Please detail the quantity of requested submissions SCE has received from EVSE manufacturers or related entities for approval per the PEV Submetering Protocol.²

c. Please detail the quantity of submeters SCE has approved per the PEV Submetering Protocol.

d. Please detail the quantity of customer submetering requests SCE has received per the PEV Submetering Protocol.

i. Please detail how many of these request have been from bidirectional charging customers.

e. Please detail the quantity of customer submetering requests SCE has approved per the PEV Submetering Protocol.

i. Please detail how many of these requests have been from bidirectional charging customers.

Response to Question 01.b-e:

(1.b) SCE has received and approved 28 EVSE embedded submeters and 2 non-embedded submeters. The full list can be found at www.sce.com/apl.

(1.c) SCE has approved 28 EVSE embedded submeters and 2 non-embedded submeters. The full list can be found at www.sce.com/apl.

(1.d) Participation in the Submeter Billing [Submetering Protocol] program requires that interested customers partner with a Meter Data Management Agent (MDMA). The MDMA's will be responsible for obtaining, verifying, and sending the submeter recorded usage data to the utility. To date, SCE has received only two MDMA applications, and has recently approved the first MDMA for participation in the program.

¹ SCE-04A, pages 130-131. <https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7663/538693971.pdf>

² D.22-08-024 and D.22-08-024: Attachment A.

<https://docs.cpuc.ca.gov/SearchRes.aspx?DocFormat=ALL&DocID=496419890>

Only approved MDMAs can submit customer applications for participation and will be able to do so for either unidirectional or bidirectional charging - that is not MDMA specific – but SCE has not yet received any customer applications for the program for any type of charging from any MDMA.

(1.e) SCE has not received customer applications for any type of charging from an approved MDMA, and therefore there are no approved customers participating in any EV submetering at this time in the program.

Southern California Edison
A.24-03-019 – 2025 SCE GRC Phase 2

DATA REQUEST SET V G I C - S C E - 0 0 1

To: VGIC-Vehicle-Grid Integration Council

Prepared by: Blake Evan Heidenreich

Job Title: Advisor

Received Date: 12/27/2024

Response Date: 1/7/2025

Question 02.a-e:

In SCE-04A, SCE notes that avoided generation capacity costs and avoided generation energy costs are to be included in the VGRP rate design, and that avoided distribution capacity costs and transmission costs are not to be included in the VGRP rate design.³

- a. Please detail the quantity of NBT exporting customers with rooftop solar in SCE's service territory currently eligible to receive both avoided generation capacity and avoided distribution capacity components for net-exported kWh.
- b. Please detail the quantity of NBT exporting customers with rooftop solar and stationary energy storage in SCE's service territory currently eligible to receive both avoided generation capacity and avoided distribution capacity components for net-exported kWh?
- c. Please provide any worksheets or information supporting SCE's proposal to offer VGRP customers avoided generation capacity costs.
- d. Please provide any worksheets or information supporting SCE's proposal to not offer VGRP customers avoided distribution and transmission capacity costs.
- e. Please provide an explanation detailing why SCE has proposed to offer avoided generation capacity costs but not avoided distribution costs to VGRP customers.

Response to Question 02.a-e:

- a. The quantity of NBT exporting customers with rooftop solar in SCE's service territory currently eligible to receive both avoided generation capacity and avoided distribution capacity components for net-exported kWh is 5,803 customers as of 12/17/2024.
- b. The quantity of NBT exporting customers with rooftop solar and stationary energy storage in SCE's service territory currently eligible to receive both avoided generation capacity and avoided distribution capacity components for net-exported kWh is 35,645 customers as of 12/17/2024.
- c. SCE's proposal to offer VGRP customers avoided generation capacity costs is intended to

³SCE-04A, page 132

<https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7663/538693971.pdf>

provide the EV owner a reasonable basis of retail export compensation that is tied to the values established by the ACC, not the retail rate, and is based on the notion that the EVs can charge away-from-site with no visibility to the pricing structure (charge-based or free) where the EV was charged. While this may not seem specifically relevant in terms of the billing compensation mechanism for VGRP, it becomes important from a cost-shift and revenue-shift perspective with increasing EV adoption, and by extension the increased potential of disparate charging behaviors across consumer groups and rate groups. SCE intends for the VGRP structure to evolve from this first step over time with increased learnings on how to value EV exports to the grid, but the first step of designing the VGRP compensation structure relying on an established tool – the ACC – based on system level avoided costs is reasonable and sufficient.

- d. SCE's proposal to not offer VGRP customers avoided distribution and transmission capacity costs results from the inability to incorporate EV exports into the planning of the grid and T&D system. It is of interest to SCE, and an objective of upcoming SCE VGI activity proposals, to determine how and to what level EVs may be integrated into T&D system planning. The ability to confidently predict energy generated from stationary BTM resources, which do not have an additional mobility function for the customer, allows SCE to reliably incorporate such BTM resources more predictably into the T&D planning process – this allows for the potential of infrastructure cost savings. Due to the evolving nature of unidirectional EV systems charging from a distributed network on the grid, and bidirectional EV systems discharging on a distributed network to the grid, the unpredictability of when EV exports may be available (i.e. the choice each customer has, daily, to export or not), the lack of a track record for EV exports, and the fact that EVs are first a means for mobility and not for grid services, SCE cannot currently reliably incorporate EV exports into its system planning. Therefore, the T&D components of the ACC were not included in the compensation value for this initial iteration of VGRP. As SCE gains experience with EV exports, it may become appropriate to include T&D elements of the ACC in future versions of VGRP.
- e. SCE has proposed to offer avoided generation capacity costs but not avoided distribution costs to VGRP customers for the reasons noted in responses 2.c and 2.d above.

Southern California Edison
A.24-03-019 – 2025 SCE GRC Phase 2

DATA REQUEST SET VGIC - SCE - 001

To: VGIC-Vehicle-Grid Integration Council

Prepared by: Blake Evan Heidenreich

Job Title: Advisor

Received Date: 12/27/2024

Response Date: 1/7/2025

Question 03.a-c:

In SCE-04A, as well as in response to CalAdvocates' SCE-DR-08, -016, and -026, SCE notes the dynamic rate options available to certain customers.⁴

a. Please provide the quantity of bidirectional charging systems enrolled in the first phase of SCE's Dynamic Rate Pilot.⁵

i. Please include what share of these customers are residential vs non-residential.

b. Please explain whether SCE has contracted with (or is in contracting with) any Automated Service Providers with experience aggregating bidirectional charging customers for SCE's Expanded Dynamic Rate Pilot.⁶

c. Please provide the quantity of bidirectional charging systems enrolled in SCE's Expanded Dynamic Rate Pilot.

i. Please include what share of these customers are residential vs non-residential.

Response to Question 03.a-c:

- a. There are no bidirectional charging systems enrolled in the first phase of SCE's Dynamic Rate Pilot.
 - a. Residential vs non-residential: NA
- b. SCE is currently in the contracting phase with Automated Service Providers (ASP) for SCE's Expanded Dynamic Rate Pilot (DRP). Both HVAC and EVs are devices with high priority for SCE to include in the Expanded DRP. One or more ASPs that is in the contracting phase has experience aggregating bidirectional charging for customers.
- c. There are currently no bidirectional charging systems enrolled in SCE's Expanded Dynamic Rate Pilot. The Expanded DRP anticipates enrolling new customers onto the pilot in 2025.
 - a. Residential vs non-residential: NA

⁴ See SCE-04A, page 129. <https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7663/538693971.pdf> and *Prepared Testimony of Additional Rate Design Proposals in SCE's 2025 GRC Phase 2*. Public Advocates Office. November 22, 2024. *Appendix B: Supporting Attachments*. B-3, B-4, B-7, B-8, and B-11 - B-15.
⁵ D.21-12-015 authorized the first phase of SCE's Dynamic Rate Pilot.